

Exam Date & Time: 12-Jun-2025 (02:00 PM - 05:00 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester Makeup Examination, June -2025

Pattern Recognition [CSE-3274]

Marks: 50

Duration: 180 mins.

### PART-A

Answer all the questions.

- 1) Compare and contrast DBSCAN with k-means in terms of:  
Shape of clusters and Sensitivity to noise. Explain with a suitable example. (5)
  - A)
  - B) Analyse the concepts of weight initialization and learning rate in the training of a backpropagation neural network. Show how weights are updated in backpropagation and what is the purpose of It? (5)
- 2) Apply the Expectation-Maximization algorithm to estimate the parameters of a Gaussian Mixture Model. (5)
  - A)
  - B) Analyse the role of learning rate and activation function in the training process of a Back Propagation Neural Network. (5)
- 3) Summarize probabilistic separability-based criterion functions. How do they help in selecting discriminative features? (3)
  - A)
  - B) Apply the concept of data abstraction in pattern recognition by describing a real-world example where it helps simplify complex data. (2)
  - C) Determine the DBSCAN algorithm and apply clustering with the following parameters:  
A(1, 2), B(2, 2), C(2, 3), D(8, 7), E(8, 8), F(25, 80), G(24, 79), H(26, 81), I(2, 1), J(1, 1)  
Parameter: Epsilon ( $\epsilon$ ) = 2, MinPts = 3 (5)
    - i. Identify the core points, border points, and noise points.
    - ii. Form the clusters based on the parameters. List of which points belong to each cluster.
- 4) Model a real-world problem using an HMM. Discuss the states, observations, and probability matrices. (5)
  - A)
  - B) Compare the concept of Gini and Entropy to determine the best attribute to split the root node of the following dataset. Show your calculations clearly. (5)

- 5) Identify and discuss the effect of cluster numbers, Cluster quality and Cluster tendency in the context of pattern recognition. (5)
- A)
- B) Compare different feature scaling methods for data preprocessing. (3)
- C) Illustrate the difference between k Means and K medoid clustering. (2)

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